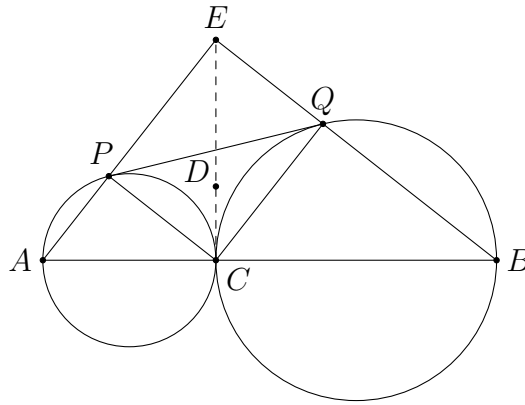


## Radical Axes — Solutions

**Problem 1.** (Sri Lanka Mathematics Challenge Competition 2022 Problem 4) A point  $C$  lies on a line segment  $AB$  between  $A$  and  $B$  and circles are drawn having  $AC$  and  $CB$  as diameters. A common tangent line to both circles touches the circle with  $AC$  as diameter at  $P \neq C$  and the circle with  $CB$  as diameter at  $Q \neq C$ . Prove that lines  $AP$ ,  $BQ$  and the common tangent line to both circles at  $C$  all meet at a single point which lies on the circle with  $AB$  as a diameter.

**Solution.**



Let  $D$  be a point inside  $\triangle CPQ$  such that  $CD$  is tangent to the two circles. Let

$$\begin{aligned} x &= \angle DCP = \angle CAP = \angle CPQ, \\ y &= \angle DCQ = \angle CBQ = \angle CQP. \end{aligned}$$

By considering the internal angles of  $\triangle CPQ$ , we obtain  $2x + 2y = 180^\circ$ , and hence  $x + y = 90^\circ$ . Note that  $ACB \perp CD$ . Thus, we also have  $\angle BCQ = 90^\circ - y = x$  and  $\angle ACP = 90^\circ - x = y$ . As  $PQ$  is tangent to  $(BCQ)$ , we have

$$\angle PQB = 180^\circ - \angle BCQ = 180^\circ - \angle BAP.$$

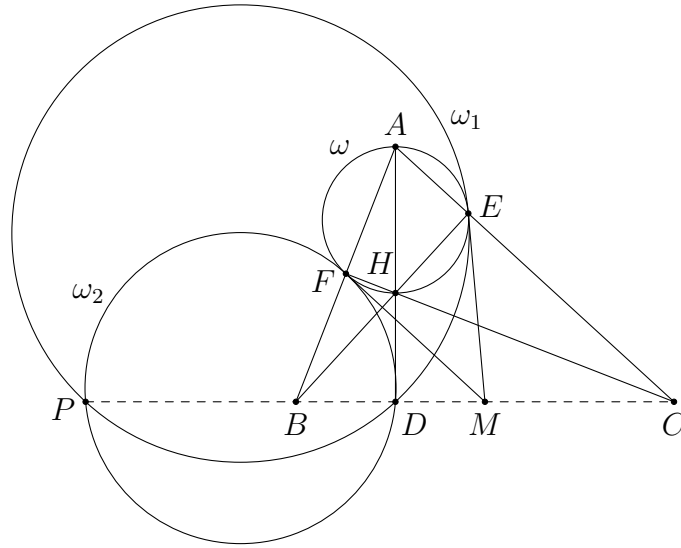
Therefore,  $B, A, P, Q$  are concyclic. It follows that  $E = AP \cap BQ$  must lie on the radical axis  $CD$  of the two circles. Lastly, since

$$\angle AEB = 180^\circ - \angle BAP - \angle ABQ = 180^\circ - x - y = 90^\circ,$$

the point  $E$  lies on the circle with diameter  $AB$ . This completes the proof. □

**Problem 2.** In acute triangle  $ABC$ , let  $D, E, F$  denote the feet of the altitudes from  $A, B, C$ , respectively, and let  $\omega$  be the circumcircle of  $\triangle AEF$ . Let  $\omega_1$  and  $\omega_2$  be the circles through  $D$  tangent to  $\omega$  at  $E$  and  $F$ , respectively. Show that  $\omega_1$  and  $\omega_2$  meet at a point  $P$  on  $BC$  other than  $D$ .

**Solution.**



Let  $M$  be the midpoint of  $BC$ . It is well-known that  $ME$  and  $MF$  are tangent to  $\omega$  (for example,  $\angle MEF = 180^\circ - \angle AEF - \angle MEC = 180^\circ - B - C = A$ ). So  $ME$  and  $MF$  are tangent to  $\omega_1$  and  $\omega_2$  respectively. As  $ME = MF$ , the power of  $M$  with respect to  $\omega_1$  and  $\omega_2$  is the same.

Note that  $M \neq D$ , since otherwise there is no circle through  $D$  tangent to  $\omega$  at  $E$  as  $DE$  would be tangent to  $\omega$ . Also,  $\omega_1$  and  $\omega_2$  cannot be the same circle since  $E \neq F$ . Therefore,  $DM$  is the radical axis of  $\omega_1$  and  $\omega_2$ . We can also show that  $\omega_1$  and  $\omega_2$  cannot be tangent at  $D$ . Indeed, if they are tangent at  $D$ , then  $DM$  is their common tangent, and so their centres must lie on  $AD$ . But then the radical axis  $ME$  of  $\omega$  and  $\omega_1$  is perpendicular to  $AD$  as the centre of  $\omega$  is the midpoint of  $AH$  where  $H$  is the orthocentre of  $\triangle ABC$ . This is clearly impossible. Thus,  $\omega_1$  and  $\omega_2$  must meet at another point  $P \neq D$  on  $BC$ .  $\square$

**Problem 3.** In a convex quadrilateral  $ABCD$ ,  $\angle ABC = \angle ADC < 90^\circ$ . Let  $E$  be a point on the side  $AB$  such that  $CB = CE$ . Let  $F$  be a point on the side  $AD$  such that  $CD = CF$ . Prove that  $C$ , the circumcentre of  $\triangle ABD$  and the circumcentre of  $\triangle CEF$  are collinear.

**Solution.** Suppose the lines  $DA$  and  $CE$  meet at  $P$ , and the lines  $BA$  and  $CF$  meet at  $Q$ . As

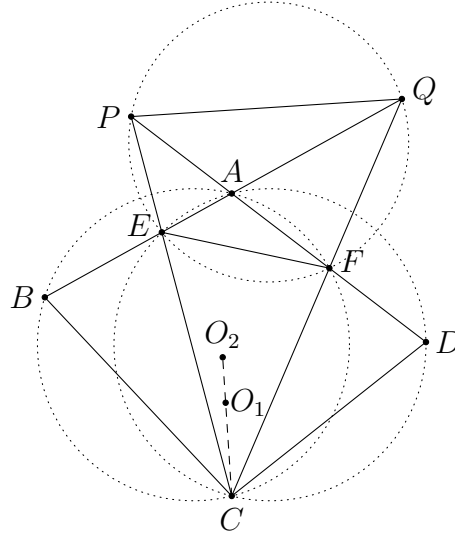
$$\angle CDA = \angle CBA = \angle CEB,$$

the points  $C, D, A, E$  are concyclic. This implies  $PA \times PD = PE \times PC$ , and hence  $P$  has the same power with respect to  $(ABD)$  and  $(CEF)$ . Similarly,  $Q$  has the same power with respect to  $(ABD)$  and  $(CEF)$ . It follows that  $PQ$  is the radical axis of the two circles.

Let  $O_1$  and  $O_2$  be the circumcentres of  $\triangle ABD$  and  $\triangle CEF$  respectively. Then we have  $PQ \perp O_1O_2$  by the property of radical axes. Now, since  $\angle PEQ = \angle CDA = \angle CFD = \angle PFQ$ , the points  $E, F, Q, P$  are concyclic. Therefore, we have

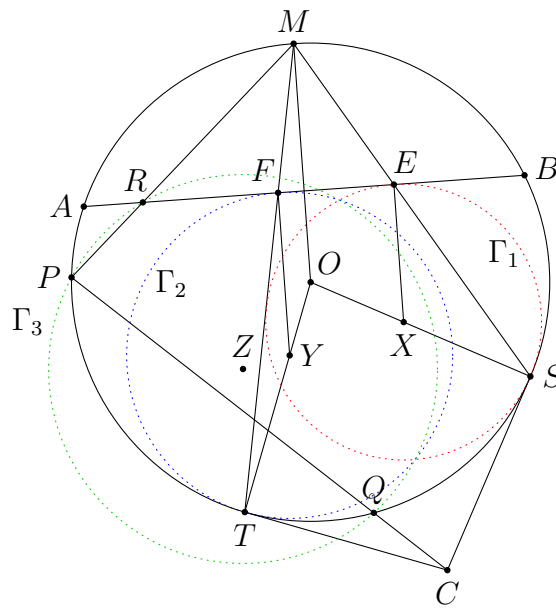
$$\angle FCO_2 = 90^\circ - \angle CEF = 90^\circ - \angle CQP,$$

which shows that  $PQ \perp CO_2$ . Thus,  $C, O_1, O_2$  are collinear. □



**Problem 4.** (China TST 2007 Test 1 Problem 1) Suppose  $AB$  is a chord of  $\odot O$ . Let  $M$  be the midpoint of arc  $AB$ . Given an arbitrary point  $C$  outside  $\odot O$ , construct tangents  $CS$  and  $CT$  to  $\odot O$  where  $S$  and  $T$  are the contact points. The lines  $MS$  and  $MT$  meet  $AB$  at  $E$  and  $F$  respectively. The lines through  $E$  and  $F$  perpendicular to  $AB$  intersect  $OS$  and  $OT$  at  $X$  and  $Y$  respectively. A line through  $C$  intersects  $\odot O$  at two distinct points  $P$  and  $Q$ . The line  $MP$  meets  $AB$  at  $R$ . Let  $Z$  be the circumcentre of  $\triangle PQR$ . Prove that  $X, Y, Z$  are collinear.

**Solution.**

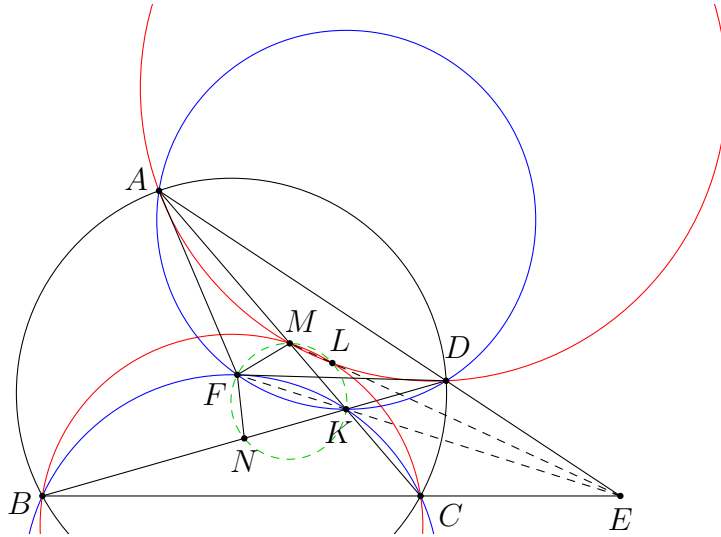


Note that  $XE \parallel OM$ . Therefore,  $\angle XES = \angle OMS = \angle OSM = \angle XSE$ , and hence  $XE = XS$ . Let  $\Gamma_1$  be the circle with centre  $X$  and passing through  $S$  and  $E$ . Similarly, let  $\Gamma_2$  be the circle with centre  $Y$  and passing through  $T$  and  $F$ . Let  $\Gamma_3$  be the circumcircle of  $\triangle PQR$ .

By the shooting lemma, we have  $ME \times MS = MF \times MT = MR \times MP$ . Therefore,  $M$  has equal power with respect to  $\Gamma_1, \Gamma_2, \Gamma_3$ . Next, note that  $\Gamma_1$  is tangent to  $\odot O$  because  $S, X, O$  are collinear. Thus,  $CS$  is tangent to  $\Gamma_1$ . Similarly,  $CT$  is tangent to  $\Gamma_2$ . Since  $CS^2 = CT^2 = CP \times CQ$ ,  $C$  has equal power with respect to  $\Gamma_1, \Gamma_2, \Gamma_3$ . It follows that  $\Gamma_1, \Gamma_2, \Gamma_3$  are coaxial with  $CM$  as the common radical axis. Thus, the centres  $X, Y, Z$  of these circles are collinear.  $\square$

**Problem 5.** (International Zhaitykov Olympiad 2011 Problem 6) Diagonals of a cyclic quadrilateral  $ABCD$  intersect at point  $K$ . The midpoints of diagonals  $AC$  and  $BD$  are  $M$  and  $N$ , respectively. The circumscribed circles  $ADM$  and  $BCM$  intersect at points  $M$  and  $L$ . Prove that the points  $K, L, M$ , and  $N$  lie on a circle. (All points are supposed to be different.)

**Solution.**



Note that  $AD$  and  $BC$  are not parallel, since otherwise  $L = N$  by symmetry. Let  $E$  be the intersection point of  $AD$  and  $BC$ . Since  $ADLM$ ,  $BCLM$  and  $ABCD$  are cyclic,  $L, M, E$  are collinear by radical axis.

Let  $F$  be the second intersection point of  $(ADK)$  and  $(BCK)$ . Since  $ADKF$ ,  $BCKF$  and  $ABCD$  are cyclic,  $F, K, E$  are collinear. It follows that

$$EK \times EF = ED \times EA = EM \times EL.$$

Therefore,  $K, F, M, L$  are concyclic.

Next, note that  $\triangle FAC \sim \triangle FDB$  by spiral similarity. As  $M$  and  $N$  are the midpoints of the corresponding sides  $AC$  and  $DB$  respectively, we have  $\triangle FAM \sim \triangle FDN$ . Therefore, we obtain  $\angle FMA = \angle FND$ , which implies  $F, N, K, M$  are concyclic. This shows  $F, N, K, L, M$  are concyclic as desired.  $\square$